

The renowned theoretical physicist Stephen Hawking used mathematics to reveal the nature of “black holes” and to explore how the Universe began. He was also a gifted science popularizer who enabled millions around the world to deepen their understanding of space-time, gravity, and humanity’s place in the cosmos.

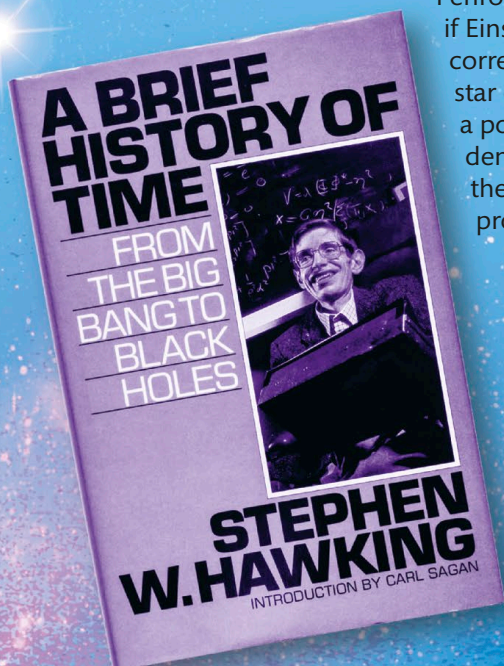
Stephen William Hawking grew up in the town of St. Albans, north of London. In 1963, while at Cambridge University studying for a doctorate in astronomy, he was diagnosed with ALS, a form of incurable motor neurone disease. Although his mental faculties were unaffected, Hawking was confined to a wheelchair from 1969. In 1985, the disease robbed him of the use of his voice, and thereafter he communicated via a computer linked to a speech synthesizer, which he controlled using his facial muscles.

Mind over matter

The term “black hole” was coined in 1967, although a German physicist, Karl Schwarzschild, had hypothesized their existence in the 1910s. It was thought that as matter is dragged into a black hole, X-rays are created, and in 1964, astronomers were able to detect “X-ray binaries” close to star-forming regions in the Milky Way, indicating a possible black hole.

While at Cambridge, Hawking became fascinated with black holes, not least because he thought that they could yield insights into the very beginning of the Universe. He studied the work of mathematician Roger Penrose (see p.300), which proved that if Einstein’s general theory of relativity is correct, the death and collapse of a giant star could culminate in a “singularity”—a point in space-time that is infinitely dense and infinitely small. In his PhD thesis, Hawking suggested that this process could run in reverse, and that

Hawking’s first popular science book attempted to make theories about the past, present, and future of the Universe accessible to nonspecialists. It was a commercial and critical success, taking him and his work into the mainstream.



MILESTONES

WRITES PhD THESIS

In “Properties of expanding universes,” written in 1966, he states that the Universe started from a singularity.

BLACK HOLES

Proves that gravitational singularities may exist while working with Roger Penrose in 1970.

KEY DISCOVERY

Draws on quantum theory in 1974 to predict that black holes emit heat and eventually evaporate.

RECEIVES HONOR

Becomes Cambridge’s Lucasian Professor of Mathematics in 1979, a chair once held by Isaac Newton.

COMMUNICATES WORK

Publishes *A Brief History of Time* in 1988. It is an instant hit and launches him to cultural stardom.